AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T316
Advanced Networking

Final Examination
Spring Semester 2025/2026

Date: 02 /June /2026

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Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 3 hours

Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

PART 1: Multiple Choice questions (10 x 2 = 20 Marks)

Question: Choose the correct Answer.

1. Which of the following wireless technologies is designed specifically for low-power, short-range communication in IoT sensor networks?
   a) 5G b) Wi-Fi c) ZigBee d) Fiber Optics
2. What is the primary characteristic of 5G networks compared to 4G?
   a) Higher latency b) Lower data rates
   c) Lower latency and higher data rates d) Decreased network capacity
3. Which protocol is the standard for secure, encrypted remote access to a network device's command line?
   a) HTTP b) SSH c) FTP d) SMTP
4. In SDN, what is the main benefit of decoupling the control plane from the data plane?
   a) Increased hardware dependency b) Manual configuration of every switch
   c) Programmatic and centralized network control d) Slower network upgrades
5. What is the main purpose of Network Address Translation (NAT)?
   a) To encrypt all web traffic b) To conserve public IPv4 addresses
   c) To replace the need for routers d) To physically connect two different buildings
6. Which technology allows voice calls to be transmitted over data networks using the Internet Protocol?
   a) VoIP b) Bluetooth c) NFC d) SON
7. In an edge computing architecture, where is the data processed to reduce latency?
   a) In a centralized remote data center b) On a nearby edge device or local server
   c) Exclusively in the public cloud d) Only on the backbone fiber cables
8. Which type of fiber optic cable is best suited for long-distance transmissions (over 10km)?
   a) Multi-mode fiber b) Coaxial cable c) Single-mode fiber d) Twisted pair
9. Network virtualization helps organizations by:
   a) Increasing the amount of physical hardware needed
   b) Reducing operational costs and improving scalability
   c) Making the network harder to manage
   d) Eliminating the need for IP addresses
10. A key capability of a "Self-Healing" network in a SON environment is:
    a) Requiring manual restarts for every failure
    b) Automatic detection and resolution of network failures
    c) Increasing the power consumption of base stations
    d) Disabling security protocols during peak hours

PART 2: Short Answer Questions (30 Marks)

Question 1 (10 marks)

A) In a multi-homed network scenario where a customer is connected to two different ISPs, explain why External BGP (eBGP) is used rather than Interior Gateway

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Protocols like OSPF for the connection between the customer and the ISPs. (5 Marks)

B) List and briefly describe two BGP settings or attributes that an administrator can use to control which ISP is preferred for outgoing traffic. (5 Marks)

Question 2 (10 marks)

A) Explain the concept of Spatial Diversity in MIMO systems and how it improves the reliability of a wireless connection in an indoor environment with many walls. (4 Marks)

B) Explain the difference between wavelength-division multiplexing (WDM) and time-division multiplexing (TDM) in optical communication systems. Provide the advantages of each technique. (6 Marks)

Question 3 (10 marks)

A) Describe the role of the SDN Controller in a large-scale network. Why is it considered the "brain" of the network? (5 Marks)

B) Explain the concept of Network Segmentation. How does this technique protect sensitive data, such as financial or patient records, from being accessed by unauthorized users on the same physical network? (5 Marks)

PART 3: Long/Problem Solving Questions (50 Marks)

Question 1: (15 marks)

A) Compare Infrastructure Mode and Ad-hoc Mode in wireless LANs. In which mode is a Wireless Access Point (WAP) required? (6 Marks)

B) Explain the CSMA/CA mechanism. Why is "Collision Avoidance" used in wireless networks instead of the "Collision Detection" (CSMA/CD) used in wired Ethernet? (5 Marks)

C) What is the purpose of the RTS/CTS (Request to Send / Clear to Send) exchange, and how does it help solve the "Hidden Node" problem? (4 Marks)

Question 2: (20 marks)

A) What are the differences between the following types of wireless channel impairments: path loss, multipath propagation, and interference from other sources? (6 marks)

B) Discuss the factors that influence indoor wireless signal propagation in modern buildings. (7 marks)

C) Explain how adaptive modulation techniques help overcome wireless communication challenges (7 Marks)

Question 3: (15 marks)

A) Define Heterogeneous Networks (HetNets) and explain their composition in terms of cell types and radio access technologies (RATs). (6 marks)

B) Discuss the major challenges faced in implementing HetNets. (4 marks)

C) How do HetNets contribute to improving overall customer experience in mobile communications? (5 marks)

End of questions

YOU EXPT